

HBLPAY

Integration Guide

ABSTRACT

The purpose of this document is to guide the online ecommerce merchant to integrate with HBL's payment gateway HBLPay. Started with a brief overview of the service followed by the pre-requisites of the technical integration. The integration section describes the step by step procedure of initiating payment. HBL also offers plugins for renowned ecommerce platform; setup details of woo-commerce and magento are given in the Plugin Support section. Annexure of the document covers sample codes, test data and response code list.

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HBLPay Overview

HBLPay is simple checkout platform that enables ecommerce merchants to accept online payments from its VISA, Master and Union Pay credit/debit cards customers. Besides this it also enables cardless payment through direct through HBL account. HBLPay is an efficient, secure and a single platform for multiple schemes.

Pre-Requisites

Your web site must meet the following requirements:

- It must have a shopping –cart, customer order creation software, or an application for initiating disbursement to send funds to payment accounts.
- The IT infrastructure must be Public Key infrastructure (PKI) enabled to use SSL-based form POST submission.
- The IT infrastructure must be capable of digitally signing customer data prior to submission to HBLPAY.

Integration

Step 1: Merchant Configuration to access HBLPay

HBL team will share the following detail with merchant in order to integrate with HBLPay.

- Merchant MID
- Merchant Password
- HBL Public Key in XML/PEM format for Encryption (RSA 4096)

Merchant technical team will provide following detail to HBL team.

- Merchant will share their public key (RSA 4096) with IPG team
- Key generation process can be found in Step II

Note: A public key is a large value that is used to encrypt data. As per our system it must be RSA 4096.

Step 2: RSA Key Generation Process

Merchant can generate RSA Key using .net library **Cryptography**. Besides that, there are many online tool available over the internet for RSA 4096 key generation and encryption assistance.

Sample Key Generation online tool URL: <https://travistidwell.com/jsencrypt/demo/> (Please be noted that it's an example and there are many more available over the internet).

XML Format

<RSAKeyValue>

```
<Modulus>3wE6KkfUfW5ptmqA6pNGnNQ59HZ8uRK3T5eq4e4Vh1a1c+RLb6zVQIrCAeTbMN5AeM9Drq3VP4r5cU5B
9h+7X1RE09i8E1a2+qgRpoB5W1PjCFaAkm/BEYHMcIeCv92aPgNB7/kxXgCksUsh6KdNQugFpxARJPOEKA1ue3m6f
AZMyX3DsrEXoMki9/XsOITXNR4BHM8+DvdgGItKuHgP4J9XPqD1C65yhKLEWSWSAHP0yRRhghLFCKrNFePXZxefFG
gGVF4Y5gqELbm8Y2Mjw6lWfwkzWRzWExHs7D26QH70Y04hG0d8mvJF71QbrNuCwtH7t712VYu1jMssG3nHxqJ5aZT
nv0kWZUp7x9ioYB1RGUMfjFdQ32QUWPg2fUbg1Un0+y0GAD3Bx6CbSnqXVwQQLoNDsXpVDgSEGkb1xJNx11rXLNcL
4MPx//ep21UQU26kZdghkFUHaGt9+iddW05vbt9gm9AUCximarHF4S5W/anCp86FE8faSJqBZvbBNQnSNu40WnVHN
20Z9Wgin8DSao4GKgHJG0TgE3iu+UrdLxbC0gF8G+zF/+doJP1n6cj09K4foV4VAiFkRYZcFUP0EmDUIWXU5Xcp+M
XGiZnKoFrozG8V3IPHHyk4RBP7Ubwv+TCZgfvrq0a8vH7EmLMQ28785qtEVNTWQn8BhtE=</Modulus>
<Exponent>AQAB</Exponent>
</RSAKeyValue>
```

PEM Format

-----BEGIN PUBLIC KEY-----

```
MIICljANBgkqhkiG9w0BAQEFAAOCAg8AMIICGgKCAgEA3wE6KkfUfW5ptmqA6pNG
nNQ59HZ8uRK3T5eq4e4Vhlalc+RLb6zVQlrCAeTbMN5AeM9Drq3VP4r5cU5B9h+7
XIREO9i8E1a2+qgRpoB5W1PjCFaAkm/BEYHMcIeCv92aPgNB7/kxXgCksUsH6KdN
QugFpxARJPOEKA1ue3m6fAZMyX3DsrEXoMki9/XsOITXNR4BHM8+DvdgGltKuHgP
4J9XPqD1C65yhKLEWSWSAHpOyRRhghLFCKrNFePXZxefFGgGVF4Y5gqELbm8Y2Mj
w6lWFwkzWRzWEXhs7D26QH7OY04hGOd8mvJF71QbrNuCwtH7t7l2VYuljMssG3nH
xqJ5aZTnv0kWZUp7x9ioYB1RGUMfjFdQ32QUWPg2fUbg1Un0+y0GAD3Bx6CbSnqX
VwQQLoNDsXpVDgSEGkblxJNx1rXLNcL4MPx//ep21UQU26kZdghkFUHaGt9+idd
W05vbt9gm9AUCximaRHf4S5W/anCp86FE8faSJqBZvbBNQnSNu4OWnVHN20Z9WGi
n8DSao4GKgHJG0TgE3iu+UrdLxbC0gF8G+zF/+doJP1n6cjO9K4foV4VAiFkRYZc
FUPOEmDUIWXU5Xcp+MXGiZnKoFrozG8V3IPHHYk4RBP7Ubww+TCZgfvrq0a8vH7E
mLMQ28785qtEVNTWQn8BhtECAwEAAQ==
```

-----END PUBLIC KEY-----

Step 3: Calling HBLPAY API for Authentication

Merchant will post the following detail on HBLPay API.

Sandbox Endpoint

<https://testpaymentapi.hbl.com/hblpay/api/checkout>

Production Endpoint

<https://digitalbankingportal.hbl.com/hostedcheckout/api/checkout>

HBLPAY support following scripting languages and you can download the sample script code for the scripting language that you use:

- .NET Sample Code
- PHP Sample Code

NOTE: Sample code Download link define in Annexure section

- Order Detail
- Billing & Shipping Detail

- Merchant Defined Data fields
- Merchant Detail (User Name & Password)

Authentication ID is provided by dev team to IPG team which merchant integrates in to their script code and request the required parameter including encrypted key in form of post.

Following are the Data Description posted by the merchant.

Authentication Fields:		
Fields	Description	Required / Optional
USER_ID	The authentication ID of merchant provided by HBL	Required
PASSWORD	Password provided by HBL to authenticate Merchant profile	Required
RETURN_URL	Merchant must provide a return URL which will be used to post back after transaction processing	Required
CANCEL_URL	Cancel URL will be called from HBL Pay if customer decides to cancel during the transaction	Required
CHANNEL	Merchant Channel	Required
TYPE_ID	Define channel specification (Union Pay, Cybersource)	Required
Order Summary		
Fields	Description	Required / Optional
DISCOUNT_ON_TOTAL	Discount offered on the total amount charged by the merchant	Optional
SUBTOTAL	Total amount of order placed by the customer	Required
Order Summary Description		
Fields	Description	Required / Optional
ITEM_NAME	Name of the specific item that customer placed the order	Required
QUANTITY	Quantity of specific items	Required
UNIT_PRICE	Per Unit Price of the specific item	Required
CATEGORY	Category of the specific item	Required
SUB_CATEGORY	Category of the specific item	Required
	Sub Category of the specific item	
Shipping Detail		
Fields	Description	Required / Optional
NAME	Name of the shipper	Optional
DELIEVERY_DAYS	Number of Days required by the merchant for delivery of the product	Optional

SHIPPING_COST	Cost of the shipment that will be deliver by the merchant	Optional
Shipping Details (Depends on merchant Profile)		
Fields	Description	Required / Optional
ship_to_address_city	City of the address that customer put at the time of the transaction where merchant has to deliver the purchased product	Optional
ship_to_address_country	Country of the address that customer put at the time of the transaction where merchant has to deliver the purchased product	Optional
ship_to_address_line1	Customer put the address at the time of the transaction where merchant has to deliver the purchased product	Optional
ship_to_address_postal_code	Customer put the postal code at the time of the transaction where merchant has to deliver the purchased product	Optional
ship_to_address_state	State of the address that customer put at the time of the transaction where merchant has to deliver the purchased product	Optional
ship_to_forename	Forename of the shipper or the specific customer who receive the purchased order	Optional
ship_to_phone	Contact Number of the shipper or the specific customer who receive the purchased order	
ship_to_surname	Surname of the shipper or the specific customer who receive the purchased order	
ship_to_email	Email address of the shipper or the specific customer who receive the purchased order	
Additional Data Fields		
Fields	Description	Required / Optional
ORDER_ID	Unique Identifier of specific transaction	Required
REFERENCE_NUMBER	Unique Identifier of specific transaction	Required
AMOUNT	total amount of the specific order	Required
CURRENCY	Allowed options of Currency	Required
PAYMENT_METHOD	Allowed options of Payment method	Required
BILL_TO_FORENAME	Forename of the Biller or the specific customer who placed the order	Required
BILL_TO_SURNAME	Surname of the Biller or the specific customer who placed the order	Required

BILL_TO_EMAIL	Email address of the biller or the specific customer who placed the order	Required
BILL_TO_PHONE	Contact Number of the Biller or the specific customer who placed the order	Required
BILL_TO_ADDRESS_LINE1	Country of the address that customer put at the time of the transaction	Required
BILL_TO_ADDRESS_CITY	City of the address that customer put at the time of the transaction	Required
BILL_TO_ADDRESS_STATE	State of the address that customer put at the time of the transaction	Required
BILL_TO_ADDRESS_COUNTRY	Country of the address that customer put at the time of the transaction	Required
BILL_TO_ADDRESS_POSTA	Customer put the postal code at the time of the transaction	Required
Merchant Define Data Fields		
Fields	Description	Required / Optional
MERCHANT DEFINED DATA 01	Channel of Operation	Required
MERCHANT DEFINED DATA 02	3D Secure Registration	Optional
MERCHANT DEFINED DATA 03	Product Category	Optional
MERCHANT DEFINED DATA 04	Product Name	Optional
MERCHANT DEFINED DATA 05	Previous Customer (If yes then Customer ID to be passed)	Optional
MERCHANT DEFINED DATA 06	Shipping Method	Optional
MERCHANT DEFINED DATA 07	Number Of Items Sold	Optional
MERCHANT DEFINED DATA 08	Product Shipping Country Name	Optional
MERCHANT DEFINED DATA 09	Hours Till Departure	Optional
MERCHANT DEFINED DATA 10	Flight Type	Optional
MERCHANT DEFINED DATA 11	Full Journey/ Itinerary	Optional
MERCHANT DEFINED DATA 12	3rd Party Booking	Optional
MERCHANT DEFINED DATA 13	Hotel Name	Optional
MERCHANT DEFINED DATA 14	Date of Booking	Optional
MERCHANT DEFINED DATA 15	Check In Date	Optional

MERCHANT DEFINED DATA 16	Check Out Date	Optional
MERCHANT DEFINED DATA 17	Product Type	Optional
MERCHANT DEFINED DATA 18	Customer ID/ Phone Number	Optional
MERCHANT DEFINED DATA 19	Country Of Top – up	Optional
MERCHANT DEFINED DATA 20	VIP Customer	Optional

Sample Request – JSON (for reference only, in actual the value of parameters should be encrypted)

```
{
  "USER_ID": "testmerchant",
  "PASSWORD": "hbl@1234",
  "RETURN_URL": "http://localhost:64149/Home/ThankYouPage",
  "CANCEL_URL": "http://localhost:64149/Home/FailedPurchase",
  "CHANNEL": "BOOKME_WEB",
  "TYPE_ID": "0",
  "ORDER": {
    "DISCOUNT_ON_TOTAL": "20",
    "SUBTOTAL": "600",
    "OrderSummaryDescription": [
      {
        "ITEM_NAME": "COMPUTER BOOK",
        "QUANTITY": "1",
        "UNIT_PRICE": "180",
        "OLD_PRICE": "210",
        "CATEGORY": "Test_Category",
        "SUB_CATEGORY": "Test SUB Category 1"
      },
      {
        "ITEM_NAME": "GUITAR",
        "QUANTITY": "2",
        "UNIT_PRICE": "200",
        "OLD_PRICE": null,
        "CATEGORY": "Test_Category2",
        "SUB_CATEGORY": "Test SUB Category2"
      }
    ]
  },
  "SHIPPING_DETAIL": {
    "NAME": "DHL SERVICE",
    "ICON_PATH": null,
    "DELIVERY_DAYS": "7",
    "SHIPPING_COST": "40"
  }
},
```

```

"ADDITIONAL_DATA": {
  "REFERENCE_NUMBER": "TEST123456789",
  "CUSTOMER_ID": null,
  "CURRENCY": "PKR",
  "BILL_TO_FORENAME": "John",
  "BILL_TO_SURNAME": "Doe",
  "BILL_TO_EMAIL": "null@cybersource.com",
  "BILL_TO_PHONE": "02890888888",
  "BILL_TO_ADDRESS_LINE": "1 Card Lane",
  "BILL_TO_ADDRESS_CITY": "My City",
  "BILL_TO_ADDRESS_STATE": "CA",
  "BILL_TO_ADDRESS_COUNTRY": "US",
  "BILL_TO_ADDRESS_POSTAL_CODE": "94043",
  "SHIP_TO_FORENAME": "John",
  "SHIP_TO_SURNAME": "Doe",
  "SHIP_TO_EMAIL": "null@cybersource.com",
  "SHIP_TO_PHONE": "02890888888",
  "SHIP_TO_ADDRESS_LINE": "1 Card Lane",
  "SHIP_TO_ADDRESS_CITY": "My City",
  "SHIP_TO_ADDRESS_STATE": "CA",
  "SHIP_TO_ADDRESS_COUNTRY": "US",
  "SHIP_TO_ADDRESS_POSTAL_CODE": "94043",
  "MerchantFields": {
    "MDD1": "mdd1",
    "MDD2": "mdd2",
    "MDD3": "mdd3",
    "MDD4": "mdd4",
    "MDD5": "mdd5",
    "MDD6": "mdd6",
    "MDD7": "mdd7",
    "MDD8": "mdd8",
    "MDD9": "mdd9",
    "MDD10": "mdd10",
    "MDD11": "mdd11",
    "MDD12": "mdd12",
    "MDD13": "mdd13",
    "MDD14": "mdd14",
    "MDD15": "mdd15",
    "MDD16": "mdd16",
    "MDD17": "mdd17",
    "MDD18": "mdd18",
    "MDD19": "mdd19",
    "MDD20": "mdd20"
  }
}
}

```

- All details will be validated on HBL Pay and return with Authorization ID to the merchant web page.

Step 4: HBLPay Page for Transaction Processing

- Merchant can access the HBL Pay page URL using that Authorization ID returned by HBL Pay API.
- Merchant will re-direct the user to following page

Sandbox Redirect URL

<https://testpaymentapi.hbl.com/hblpay/site/index.html#/checkout?data=>

Production Redirect URL

<https://digitalbankingportal.hbl.com/hostedcheckout/site/index.html#/checkout?data=>

In above URL, data field will contain encrypted authorization ID

- After getting landed on HBL Pay's page merchant then now able to see the checkout page along with details and multiple options of payments i-e Union Pay, Cybersource and Direct Transfer. All three platforms i-e Cybersource, Direct Transfer & Union Pay are available for online collection payments.

Card holder have the choice to pay the payment via following option.

- Visa/Master Card
- HBL Account
- UnionPay Card




TITLE

Billing Details

First Name *
John

Last Name *
Doe

Company Name (Optional)

Country *
US

Town/ City *
My City

State *
CA

Postal Zipcode *
94043

Street Address *
1 Card Lane

Phone (Optional)
2890888888

Email (Optional)
null@cybersource.c

Order Note (Optional)

Payment Options

☒ HBL Account
 ☒ Visa/Master Card
 ☐ Union Pay

Click "Submit" To Proceed Further.

[Proceed To Cybersource](#)

Session will expire in
54:24

COMPUTER BOOK	400	40
Qty : 2		
SCIENCE BOOK		200
Qty : 2		
Total		460

Shipping Details

Shipping Method	Process Days	Shipping Cost
DHL SERVICE	7	40

If customer wants to purchase the item via **Visa/Master Card**, select the option, click the button **Proceed to Visa/Master Card**.

- Put all the details (Card Type, Card number, Expiry date, CVN) on payment details page.
- Customer will redirect to the Cybersource page for payment.

Powered by
HBL InternetPaymentGateway

PaymentReviewReceipt

Payment Details

* Required field

Card Type *

☐  Visa ☐  Mastercard

Card Number *

Expiration Date *

CVN *

This code is a three or four digit number printed on the back or front of credit cards.



Next

[Cancel Order](#)

Your Order

Total amount\$460.00

- By clicking the **Next** button, customer will review all the details and then click the **Pay** button.

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Powered by
HBL InternetPaymentGateway

Payment Review Receipt

Review your Order

Payment Details		Your Order	
Card Type	Visa	Total amount	\$460.00
Card Number	xxxxxxxxxxxx0101		
Expiration Date	05-2023		

BackPay

[Cancel Order](#)

If customer wants to purchase the item via **HBL Account**, select the option, click the button **Proceed to HBL Account**.

- If customer is the HBL account holder and wants to purchase the item via **HBL Account**, select the option, put the details (i-e account number and CNIC number) and click the **Proceed** button.




TITLE

Billing Details

First Name *

Last Name *

Company Name (Optional)

Country *

Town/ City *

State *

Postal Zipcode *

Street Address *

Phone (Optional)

Email (Optional)

Order Note (Optional)

Payment Options

 HBL Account
  Visa/Master Card
  Union Pay

Account Number

CNIC

Enter 14 Digit **HBL Branch** Account No

Or

Enter 13 digit CNIC number

11 Digit **HBL Konnect** Account No

Proceed

Session will expire in

54:38

COMPUTER BOOK	400	40
Qty : 2		
SCIENCE BOOK		200
Qty : 2		
Total		460

Shipping Details

Shipping Method	Process Days	Shipping Cost
DHL SERVICE	7	40

- If customer attempt the transaction first time on merchant website, Customer will receive a verification call on registered number and after successful verification, **OTP** will send to customer via **SMS**.




TITLE

Billing Details

First Name * Last Name *

Company Name (Optional)

Country * Town/ City *

State * Postal Zipcode *

Street Address *

Phone (Optional) Email (Optional)

Order Note (Optional)

Payment Options

☒ HBL Account
 ☐ Visa/Master Card
 ☐ Union Pay

OTP

Note:
If this is your first HBL Direct Transfer Transaction, you will receive a verification call. After Successful verification, an OTP will be sent via SMS

[Proceed](#)

Session will expire in 54:37

SCIENCE BOOK	200
Qty : 2	
COMPUTER BOOK	40
Qty : 2	
Total	460

Shipping Details

Shipping Method	Process Days	Shipping Cost
DHL SERVICE	7	40

- By putting the OTP on the page, customer will **Proceed** the transaction and get the message of **Payment Successful**.

Payment Successful.

Redirecting to page 00:09 seconds, or click [url](#) to get back.

TITLE

Billing Details

First Name * Last Name *

Company Name (Optional)

Country * Town/ City *

State * Postal Zipcode *

Street Address *

Phone (Optional) Email (Optional)

Order Note (Optional)

Payment Options

☒ HBL Account
 ☐ Visa/Master Card
 ☐ Union Pay

OTP

Note:
If this is your first HBL Direct Transfer Transaction, you will receive a verification call. After Successful verification, an OTP will be sent via SMS

[Proceed](#)

Session will expire in 54:13

SCIENCE BOOK	200
Qty : 2	
COMPUTER BOOK	40
Qty : 2	
Total	460

Shipping Details

Shipping Method	Process Days	Shipping Cost
DHL SERVICE	7	40

- If customer wants to purchase the item via **Union pay Card**, select the option, click the button **Proceed to Union pay**.




TITLE

Billing Details

First Name *

John

Last Name *

Doe

Company Name (Optional)

Country *

US

Town/ City *

My City

State *

CA

Postal Zipcode *

94043

Street Address *

1 Card Lane

Phone (Optional)

2890888888

Email (Optional)

null@cybersource.c

Order Note (Optional)

Payment Options

HBL Account

 Visa/Master Card

 Union Pay

Click "Submit" To Proceed Further.

Proceed To Unionpay

Session will expire in

54:13

COMPUTER BOOK	₹80	40
Qty : 2		
SCIENCE BOOK		200
Qty : 2		
Total		460

Shipping Details

Shipping Method	Process Days	Shipping Cost
DHL SERVICE	7	40

- Put the details (Card number, Expiry date, CVN and SMS Code) on payment details page.

Order Amount: 4.60 CNY

Order Number: 224fc872ce...e4469eea Merchant Name: 测试虚拟商户77729005811... [Order details\[+\]](#)

1. Enter card NO. ➡ 2. Verify card Info. ➡ 3. Complete

[Next](#)[UnionPay Card](#) | [MiniPay](#)

Problem with making payment?

1. Why is installing ActiveX control necessary? ActiveX control wouldn't download, what should I do?

2. When making payment, what should I do when a "There is a problem with this website's security certificate!" warning appears?

- Customer will review the payment details and Click the **Confirm and Pay** button.

Order Amount: **4.60** CNY

Order Number: 224fc872ce...e4469eea

Merchant Name: 测试虚拟商户77729005811... [Order details\[+\]](#)

Card Number:  6250****0014 (CreditCard) [Choose other card](#)

Expiration Date: MM YY
Please enter expiration date of your bank card. e.g. 09/13

CVN2:
Please enter the last three digits on the back of your card

SMS Code: [Send Free SMS](#)

☒ I have acknowledged and accepted
Please confirm that you have acknowledged and accepted the agreement on this service

☒ Remember card number

[Confirm and Pay](#)

[UnionPay Card](#) | [MiniPay](#)

Problems with making payments?

1.What is a registered banking mobile number?

Answer: The registered banking mobile number is mobile number you registered with the bank while applying for bank card or processing bank card related business.

2.Why is installing ActiveX control necessary? ActiveX control wouldn't download, what should I do?

3.When making payment, what should I do when a "There is a problem with this website's security certificate!" warning appears?

- By clicking the **Confirm and Pay** customer will get the message of **Payment Successful**.

Payment Succeeded **4.60** CNY !To view merchant order status, please click [Return to Merchant](#)You can also: [Check transaction records](#)

To help in answering any further question you may have regarding this transaction, please save the merchant transaction ID: 224fc872ceaa4e1cb4d93268e4469eea

Promotion:

Plugin Support

HBLPay do offer No-Code integration model as well. If merchant website is developed on any of the following platforms, then by installing and configuring our plugin package will quickly enable merchants for accepting payments through our gateway.

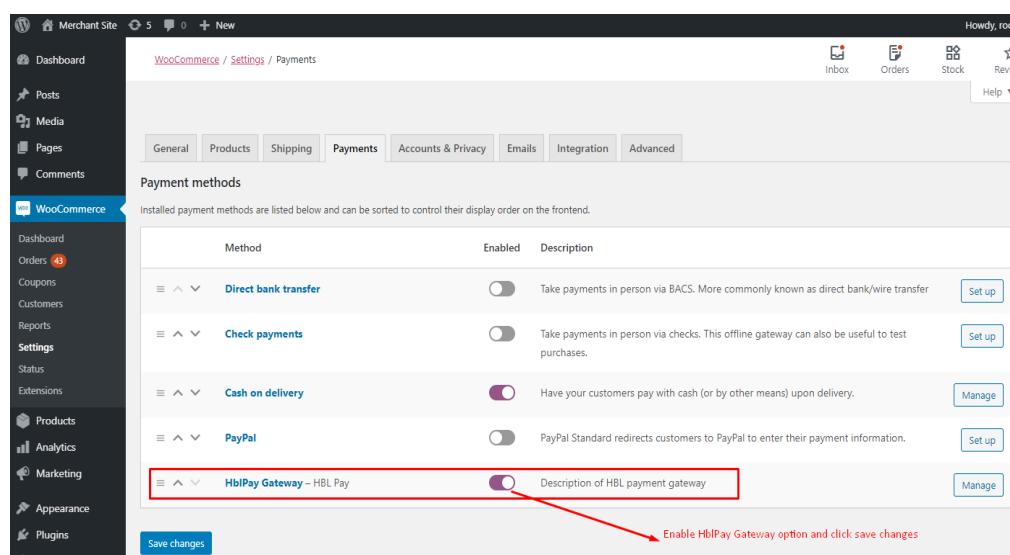
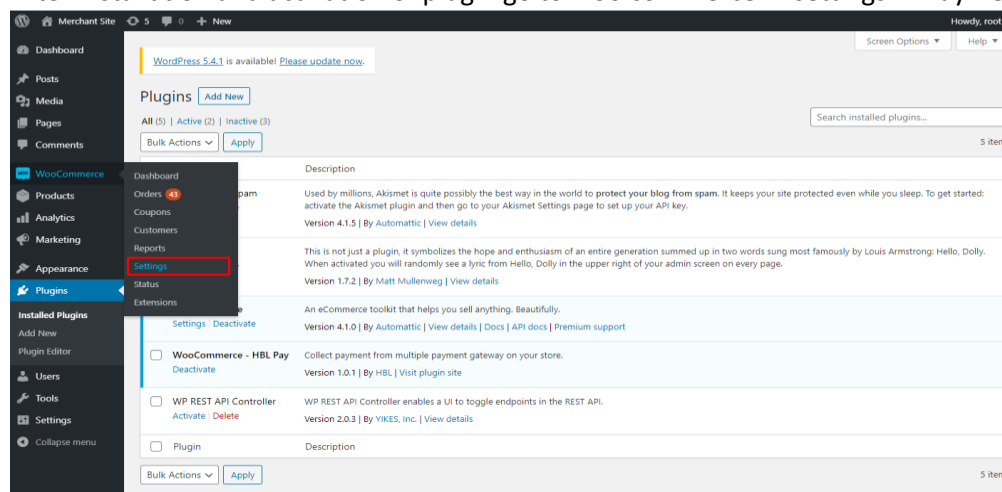
Woo-Commerce Setup

1. HBL team will provide merchant HBLPAY plugin as .zip file
2. To install plugin use the following link to install. You may proceed with the upload zip file step

<https://www.wpbeginner.com/beginners-guide/step-by-step-guide-to-install-a-wordpress-plugin-for-beginners/>



3. After installation and activation of plugin go to woo commerce -> settings -> Payments



4. After enabling the payment option go to manage and add plugin settings and save.

Magento Setup

1. Install plugin with the help of following link.

<https://bsscommerce.com/blog/how-to-extension-in-magento-2/>



[install-](#)

2. After installation and execution of commands go to admin dashboard -> Stores -> Configuration -> Sales -> Payments

3. You will find HBL Pay gateway in a list of payment methods. Enable gateway with the yes option and fill complete settings to start using HBLPay gateway.

Annexures

.Net Sample Code

```

public string CallService(string jsonData)
{
    string URL =
System.Configuration.ConfigurationSettings.AppSettings["ServiceURL"];
    string AllowProxy =
System.Configuration.ConfigurationSettings.AppSettings["AllowProxy"];
    string Proxy =
System.Configuration.ConfigurationSettings.AppSettings["Proxy"];
    string IsLive =
System.Configuration.ConfigurationSettings.AppSettings["IsLive"];
    HttpWebRequest request = (HttpWebRequest)WebRequest.Create(URL);
    WebProxy proxy = new WebProxy(Proxy, true);
    HBLPayResponseModel hblResponse = new HBLPayResponseModel();
    if (IsLive == "1")
    {
        ServicePointManager.Expect100Continue = true;
        ServicePointManager.SecurityProtocol = SecurityProtocolType.Ssl3 |
SecurityProtocolType.Tls | SecurityProtocolType.Tls11 | SecurityProtocolType.Tls12;
    }
    request.Method = "POST";
    request.ContentType = "application/json";
    if (AllowProxy == "1")
        request.Proxy = proxy;
    request.ContentLength = jsonData.Length;
    using (Stream webStream = request.GetRequestStream())
    using (StreamWriter requestWriter = new StreamWriter(webStream,
System.Text.Encoding.ASCII))
    {
        requestWriter.Write(jsonData);
    }
    try
    {
        string finalResult = "";
        WebResponse webResponse = request.GetResponse();
        using (Stream webStream = webResponse.GetResponseStream())
        {
            if (webStream != null)
            {
                using (StreamReader responseReader = new StreamReader(webStream))
                {
                    string response = responseReader.ReadToEnd();
                    finalResult = response;
                }
            }
        }
        hblResponse =
JsonConvert.DeserializeObject<HBLPayResponseModel>(finalResult);
        if (hblResponse.Data != null &&
!string.IsNullOrEmpty(hblResponse.Data.SESSION_ID))

```

```

{
    var plainTextBytes =
        System.Text.Encoding.UTF8.GetBytes(hblResponse.Data.SESSION_ID);
    hblResponse.Data.SESSION_ID =
        System.Convert.ToBase64String(plainTextBytes);
}
finalResult = JsonConvert.SerializeObject(hblResponse);
return finalResult;
}
catch (Exception ex)
{
    throw new Exception(ex.Message);
}
}

```

You will receive SESSION_ID in response from API call. Then open the URL from your .net code or JAVASCRIPT code with session id.

For example:

```

window.location.href =
    'https://testpaymentapi.hbl.com/HBLPay/Site/index.html#/checkout?data=' + sessionID;

```

PHP Sample Code

```
function callAPI($method, $url, $data)
{
    $is_live = 'no';
    use_proxy === 'no';
    $curl = curl_init();
    switch ($method){
        case "POST":
            curl_setopt($curl, CURLOPT_POST, 1);
            if ($data)
                curl_setopt($curl, CURLOPT_POSTFIELDS, $data);
            break;
        case "PUT":
            curl_setopt($curl, CURLOPT_CUSTOMREQUEST, "PUT");
            if ($data)
                curl_setopt($curl, CURLOPT_POSTFIELDS, $data);
            break;
        default:
            if ($data)
                $url = sprintf("%s?%s", $url, http_build_query($data));
    }
    // OPTIONS:
    curl_setopt($curl, CURLOPT_URL, $url);
    curl_setopt($curl, CURLOPT_HTTPHEADER, array('Content-Type: application/json'));
    curl_setopt($curl, CURLOPT_RETURNTRANSFER, 1);
    curl_setopt($curl, CURLOPT_HTTPAUTH, CURLAUTH_BASIC);
    if($is_live === 'yes')
    {
        curl_setopt($curl, CURLOPT_SSL_VERIFYHOST, 0);
        curl_setopt($curl, CURLOPT_SSL_VERIFYPEER, 0);
    }
    // PROXY
    if($use_proxy === 'yes')
    {
        $proxy = 'your proxy';
        curl_setopt($curl, CURLOPT_PROXY, $proxy);
    }
    //EXECUTE
    $result = curl_exec($curl);
    if(!$result){die("Connection Failure");}
    curl_close($curl);
    return $result;
}
```


You will receive SESSION_ID in response from API call. Then open the URL from your .net code or JAVASCRIPT code with session id.

For example:

```
window.location.href =  
'https://testpaymentapi.hbl.com/HBLPay/Site/index.html#/checkout?data=' + sessionId;
```

Test Accounts (for Direct Transfer) **

Type	CNIC	Account #	Passcode/OTP
HBL Branch Account			In test environment OTP SMS can be viewed on below given URL
HBL Konnect Branchless Account			In test environment OTP SMS can be viewed on below given URL

** P.S. HBL Implementation team will provide Direct Transfer test data

<https://testpaymentapi.hbl.com/OTPViewer/Home/Email>

Test Cards

Scheme	Security	Card #	Expiry Month	Expiry Year	CVV	Passcode
Visa	Non 3D	4000000000000101	05	2023	111	
Visa	3D	4000000000000002	05	2023	111	1234
Master	Non 3D	5200000000000114	05	2023	111	
Master	3D	5200000000000007	05	2023	111	1234
Union Pay	Debit Card	Card #: 6223164991230014 Mobile #: 13012345678	12	33	123	PC Code: 111111 Mobile Code: 123456
Union Pay	Credit Card	Card #: 6250947000000014 Mobile #: +852 11112222	12	33	123	PC Code: 111111 Mobile Code: 123456

Response Codes

Response Code	Response Message
100	Dear Customer, Thank you for your payment, Your transaction has been received successfully.
0	Dear Customer, Thank you for your payment, Your transaction has been received successfully.
481	Dear Customer, Your transaction has been received and under review, our representative will contact you shortly for further processing.
89	Channel is NOT ALLOWED
90	SESSION ID NOT FOUND
37	TRANSACTION_CODE_NOT_FOUND
35	INVALID_TRANSACTION
34	PARTIAL_REFUND_NOT_ALLOWED
33	AMOUNT_IS_GREATER_THAN_EXSISTING_AMOUNT
38	TPE_RESPONSE_CODE_NOT_FOUND
36	TRANSACTION TYPE NOT FOUND
230	Dear Customer, Your transaction has been declined by your issuing bank, Kindly contact your issuing bank for further details.
70	Transaction has been redirected, waiting for cardholder input
76	Requires encryption verification for activation
78	Issuer transaction rights limited, for details please contact your issuer
85	Transaction failed, the marketing rules are not met
1012	Account Closed
520	Dear Customer, Your transaction has been declined by your issuing bank, Kindly contact your issuing bank for further details.
230	Dear Customer, Your transaction has been declined by your issuing bank, Kindly contact your issuing bank for further details.
42	Balance deduction successful but transaction exceeded payment time limit
11020	ACCOUNT_STATUS_NOT_FOUND
23	AMOUNT IS GREATER THAN MAX AMOUNT
21	CERTIFICATE NOT PROVIDED
01	CLIENT CERTIFICATE AUTHENTICATION FAILED
02	INVALID USERNAME OR PASSWORD
03	INVALID CNIC
05	ACCOUNT NOT FOUND
06	MOBILE NO. NOT FOUND
07	INCORRECT TXN. REFERENCE NO.
08	INCORRECT OTP
09	OTP EXPIRED
10	INSUFFICIENT FUNDS
11	SYSTEM ERROR
12	API NOT ALLOWED
13	Invalid Channel Password
14	INCORRECT AMOUNT
1111	DAILY DEBIT LIMIT EXCEED
1112	MONTHLY DEBIT LIMIT EXCEED

1113	YEARLY DEBIT LIMIT EXCEED
1114	DAILY CREDIT LIMIT EXCEED
1115	MONTHLY CREDIT LIMIT EXCEED
1116	YEARLY CREDIT LIMIT EXCEED
11008	INCORRECT PASSCODE
11004	PASSCODE IS EXPIRED
11003	PASSCODE IS VERIFIED
11005	PASSCODE IS PENDING
11006	CODE STRENGTH FAIL
11007	INVALID PASSCODE TYPE
11008	INCORRECT PASSCODE
11009	PASSCODE RETRY EXHAUSTED
11011	EXCEPTION
11010	CODE EXIST
11012	CERTIFICATE NOT PROVIDED
11013	CLIENT CERTIFICATE AUTHENTICATION FAILED
188	INVALID USERNAME OR PASSWORD
76	INVALID CNIC
174	CUSTOMER NOT FOUND
143	ACCOUNT NOT FOUND
202	MOBILE NUMBER NOT FOUND
11014	INCORRECT TRANSACTION REFERENCE NUMBER
11008	INCORRECT PASSCODE
11004	PASSCODE EXPIRED
1003	INSUFFICIENT BALANCE
1	SYSTEM ERROR
11015	API NOT ALLOWED
11016	API LIMIT EXCEED
02	INCORRECT AMOUNT
102	DAILY DEBIT LIMIT EXCEED
88	There is a special condition on your account
63	INSUFFICIENT FUNDS
11018	MONTHLY TRANSACTION COUNT LIMIT EXCEED
11019	YEARLY TRANSACTION COUNT LIMIT EXCEED
173	TRANSACTION NOT FOUND
9000	GENERAL ERROR
1110	DAILY TXN COUNT LIMIT EXCEED
1117	MONTHLY TXN COUNT LIMIT EXCEED
1118	YEARLY TXN COUNT LIMIT EXCEED
69	TRANSACTION NOT FOUND
320	TRANSACTION NOT ALLOWED ON THAT ACCOUNT TYPE
24	CNIC_BLOCKED
11017	DAILY TRANSACTION COUNT LIMIT EXCEED
04	TXN ALREADY CONSUMED
22	AMOUNT IS LESS THAN MIN AMOUNT
79	ACCOUNT NOT FOUND
50	ACCOUNT NOT FOUND

39	REVERSAL DATE EXCEED
236	Dear Customer, Your transaction has been declined by your issuing bank, Kindly contact your issuing bank for further details.
476	Dear Customer, Your transaction has been declined by your issuing bank, Kindly contact your issuing bank for further details.
106	INVALID BYPASS ID
105	Request Cancelled
99	INVALID ADDITIONAL DETAILS
110	INVALID_URL
271	Bio-metric Verification Failed
111	CURRENCY_NOT_ALLOWED
109	INVALID CUSTOMER ID
108	INVALID_ORDER_REFERENCE_NUMBER
80	Your account is not allowed to be registered for HBL InternetBanking
01	Transaction failed. For details please inquire overseas service hotline.
02	System is not started or temporarily down, please try again later
03	Transaction communication time out, please initiate inquiry transaction
05	Transaction has been successful, please inquire about transaction result shortly
06	System is busy, please retry it later.
10	Message format error
11	Verify signature error
12	Repeat transaction
13	Message transaction key element missing
30	Transaction failed, please try using other UnionPay card for payment or contact overseas service hotline
31	Merchant state incorrect. The payment is not completed within the order timeout.
32	No such transaction right
33	Transaction amount exceeds limit
34	Could not find this transaction
35	Original transaction does not exist or state is incorrect
36	Does not match original transaction information
37	Max number of inquiries exceeded or too frequent operations
38	UnionPay risk constraint
39	Transaction is not within the acceptance time range
43	Business not allowed, please contact overseas service hotline for help.
44	Wrong number entered or business not opened, please contact overseas service hotline for help.
45	The original transaction has been refunded or cancelled successfully.
60	Transaction failure, for details, please inquire with your issuer
61	Card number entered is invalid, please double check and enter
62	Transaction failed, issuer does not support this merchant, please change to another bank card
63	Card state is incorrect
64	Card balance is insufficient
65	Error with PIN, expiration date, or CVN2 entered, transaction failure
66	Cardholder identity information or mobile number entered are incorrect, verification failure
67	Limit on number of PIN entry attempts exceeded

68	Your bank card currently does not support this business, please inquire with your bank or overseas service hotline for help
69	Time limit on entry exceeded, transaction failure
71	Dynamic password or SMS verification code validation failure
72	You have not signed up for UnionPay card-not-present payment service at the bank counter or on your personal online bank, please go to a bank counter or access your online banking to activate it or contact overseas service hotline for help.
73	Payment card has exceeded expiration date
77	Bank card has not been activated for authenticated payment
79	The bank card is valid, but issuer does not support SMS verification
80	Transaction failed and the token has expired
81	Monthly accumulated transaction counter (amount) exceeded
82	PIN needs to be verified
84	PIN is required but not submitted
86	QRC status error
88	QRC not found
89	No Token found, invalid TR status or invalid Token status
98	File does not exist
99	General error
00	Dear Customer, Thank you for your payment, Your transaction has been received successfully.
112	Dear customer, you cancelled the order
218	MONTHLY DEBIT LIMIT EXCEED
100	Successful transaction
101	The request is missing one or more required fields.see the reply fields missingField_0 through missingField_N for which fields are missing
104	The merchant reference code for this authorization request matches the merchant reference code of another authorization request that you sent within the past 15 minutes.
150	General system failure.
153	Your CyberSource account is not enabled for the OCT service. Contact CyberSource Customer Support to have your account enabled for this service.
201	The issuing bank has questions about the request. You do not receive an authorization code programmatically, but you might receive one verbally by calling the processor. Possible action: call your processor to possibly receive a verbal authorization. For contact phone numbers, refer to your merchant bank information.
203	General decline of the card. No other information was provided by the issuing bank. Possible action: request a different card or other form of payment
205	Stolen or lost card
208	Inactive card or card not authorized for card-not-present transactions
231	Invalid account number
233	General decline by the processor
234	Incorrect information in your CyberSource account
240	The card type sent is invalid or does not correlate with the credit card number
250	The request was received, but a timeout occurred at the payment processor
490	Your aggregator or acquirer is not accepting transactions from you at this time
491	Your aggregator or acquirer is not accepting this transaction
107	INVALID BILLING DETAILS
91	ORDER_NOT_FOUND

92	INVALID_ORDER_DETAILS
93	INVALID_SHIPPING_DETAILS
94	SHIPPING_DETAILS_NOT_FOUND
95	SESSION_EXPIRED
96	INVALID_QUANTITY
25	INVALID CLIENT IP
26	ORIGINAL_TRANSACTION_NOT_FOUND
28	INVALID CUSTOMER KEY
29	CLIENT NOT FOUND
30	INVALID_DATA
31	CUSTOMER_ALREADY_REGISTER
97	INVALID_SHIPPING_NAME
98	DECRYPTION_FAILED
219	YEARLY DEBIT LIMIT EXCEED
46	SYSTEM ERROR
44	INTERNAL ERROR
102	Dear Customer, Your transaction has been declined by your issuing bank, Kindly contact your issuing bank for further details.
104	Dear Customer, Your transaction has been rejected because of unsuccessful payment.
110	Only a partial amount was approved
150	Dear Customer, Your transaction has been rejected because of unsuccessful payment.
151	Dear Customer, Your transaction has been rejected because of unsuccessful payment.
152	Dear Customer, Your transaction has been rejected because of unsuccessful payment.
200	Dear Customer, Your transaction has been declined by your issuing bank, Kindly contact your issuing bank for further details.
201	Dear Customer, Your transaction has been declined by your issuing bank, Kindly contact your issuing bank for further details.
202	Dear Customer, Your transaction has been declined by your issuing bank, Kindly contact your issuing bank for further details.
203	Dear Customer, Your transaction has been declined by your issuing bank, Kindly contact your issuing bank for further details.
204	Dear Customer, Your transaction has been declined by your issuing bank, Kindly contact your issuing bank for further details.
205	Dear Customer, Your transaction has been declined by your issuing bank, Kindly contact your issuing bank for further details.
207	Dear Customer, Your transaction has been declined by your issuing bank, Kindly contact your issuing bank for further details.
208	Dear Customer, Your transaction has been declined by your issuing bank, Kindly contact your issuing bank for further details.
210	Dear Customer, Your transaction has been declined by your issuing bank, Kindly contact your issuing bank for further details.
211	Dear Customer, Your transaction has been declined by your issuing bank, Kindly contact your issuing bank for further details.
221	Dear Customer, Your transaction has been declined by your issuing bank, Kindly contact your issuing bank for further details.
222	Dear Customer, Your transaction has been declined by your issuing bank, Kindly contact your issuing bank for further details.

231	Dear Customer, Your transaction has been declined by your issuing bank, Kindly contact your issuing bank for further details.
232	Dear Customer, Your transaction has been declined by your issuing bank, Kindly contact your issuing bank for further details.
233	Dear Customer, Your transaction has been declined by your issuing bank, Kindly contact your issuing bank for further details.
234	Dear Customer, Your transaction has been declined by your issuing bank, Kindly contact your issuing bank for further details.
240	Dear Customer, Your transaction has been declined by your issuing bank, Kindly contact your issuing bank for further details.
475	Dear Customer, Your transaction has been declined by your issuing bank, Kindly contact your issuing bank for further details.
520	Dear Customer, Your transaction has been declined by your issuing bank, Kindly contact your issuing bank for further details.

Customer Friendly Messages

Given below or similar messages can be shown to the customer after successful or a failed transaction.

- If transaction response / decision received as **“Accept”** or **“Successful”** please display message like below;

Dear Customer, thank you for your payment, your transaction has been received successfully and your account will be updated accordingly.

- If transaction response / decision received as **“Declined”** please display message like below:

Dear Customer, your card transaction has been declined by your card issuing bank, Kindly contact your card issuing bank for details.

- If transaction response / decision received as **“Reject”** (with Reason Code 481) please display message like below.

Dear Customer, your card transaction has been received and under review, our representative will contact you shortly for further process.